

# Beyond Missingness Rate: Structured Stress Testing and Feature Contracts for Passive Sensing Data

Tae Ho Kim | Independent Researcher

Code & reproduction  
github.com/dansojo/ictc2026-missingness-stress-test  
sinzenight@gmail.com



## 1 Introduction

Passive sensing data collected by smartphones or wearables may contain missing records due to device state or failures, or because participants do not wear or use the devices. Even at the same missingness rate, sporadic losses at multiple time points and losses over a continuous interval can distort features calculated from the remaining records differently. Evaluating missingness therefore requires considering both the missingness rate and the pattern of missing records.

## 2 Methods

For the 10 participants in the ETRI passive sensing dataset, 10 days were selected for each participant and measurement item, prioritizing dates with higher observation availability. Ties were resolved by prioritizing dates with a shorter longest continuous missing interval relative to the analysis window. The initial setting for the number of selected days was retained. Across eight measurement items, this yielded 800 participant–day–item combinations.

For contiguous missingness, an interval spanning 20% of each feature’s analysis window was placed at a randomly selected starting position, and records within that interval were removed. For random missingness, records were removed at random from the same data, with the count matched to the valid records actually removed by contiguous deletion. This was repeated 50 times for each participant–day–item combination.

Features of event count, state ratio, intensity, and timing were recalculated under both conditions. The absolute difference from the value before removal was divided by a fixed, pre-specified scale to obtain standardized error, and errors were compared between conditions. The reference was the value calculated from the originally observed records, not a true value that includes unobserved records.

### Constructing the missingness conditions

One cell = one valid observation

☐ Retained

☒ Deleted

Original records

24 before deletion

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

Contiguous deletion

K = 9 removed

Random deletion

K = 9 removed

A randomly positioned 20% time window removes K = 9 records. Random deletion removes K = 9 from the same original records, matching the actual contiguous deletion count.

All rows show the same original records in chronological order. Equal cell width and spacing show record order, not elapsed time. Hypothetical 24-record illustration, not experimental data.

## 3 Feature contracts

Table I. Contracted statistics and analysis support

|                                                       |                                                                                |                                                                |
|-------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Event count</b><br>Finite pairs<br>4,995 / 5,000   | <b>Contracted statistic</b><br>Sum of valid nonnegative event contributions    | <b>Eligibility</b><br>≥1 valid event contribution              |
| <b>State ratio</b><br>Finite pairs<br>10,000 / 10,000 | <b>Contracted statistic</b><br>Positive valid-state units / valid state units  | <b>Eligibility</b><br>Positive valid-state denominator         |
| <b>Intensity</b><br>Finite pairs<br>14,840 / 15,000   | <b>Contracted statistic</b><br>Usage duration sum; light mean log(1 + lux)     | <b>Eligibility</b><br>≥1 finite valid nonnegative contribution |
| <b>Timing</b><br>Finite pairs<br>9,790 / 10,000       | <b>Contracted statistic</b><br>90th percentile weighted by positive event mass | <b>Eligibility</b><br>≥3 positive events; total event mass > 0 |

Adapted from Table I. Finite pairs: defined-output pairs / scheduled pairs. Reference features use observed support.

## 6 Conclusion and limitations

The effect of missingness pattern on feature distortion differed across feature families, even when equal numbers of valid records were deleted. State ratio and intensity showed higher error under contiguous than random deletion, but these findings were insufficient to satisfy the overall decision criterion. The overall decision therefore remained inconclusive. In ETRI prediction tests, participant-level analysis did not establish that contiguous deletion affected predictions more than random deletion.

To examine this difference, this study removes the same number of originally observed valid records either at random or from a continuous interval. It then evaluates how much features of event count, state ratio, intensity, and timing change relative to their values before removal. A “feature contract” specifies how each feature is calculated, the observation conditions required for calculation, and when calculation is withheld if those conditions are not met. These contracts guide the evaluation of feature distortion under different missingness patterns, while predictive model stability is evaluated separately.

## 4 Primary results

We deleted a contiguous interval spanning 20% of the analysis window, then matched random deletion to the valid-record count actually removed.

**$\Delta e = e(\text{contiguous}) - e(\text{random})$**

$\Delta e > 0$  means higher standardized error under contiguous deletion, despite equal valid-record deletion counts.

$e = |f(\text{masked}) - f(\text{observed})| / \text{fixed robust scale}$

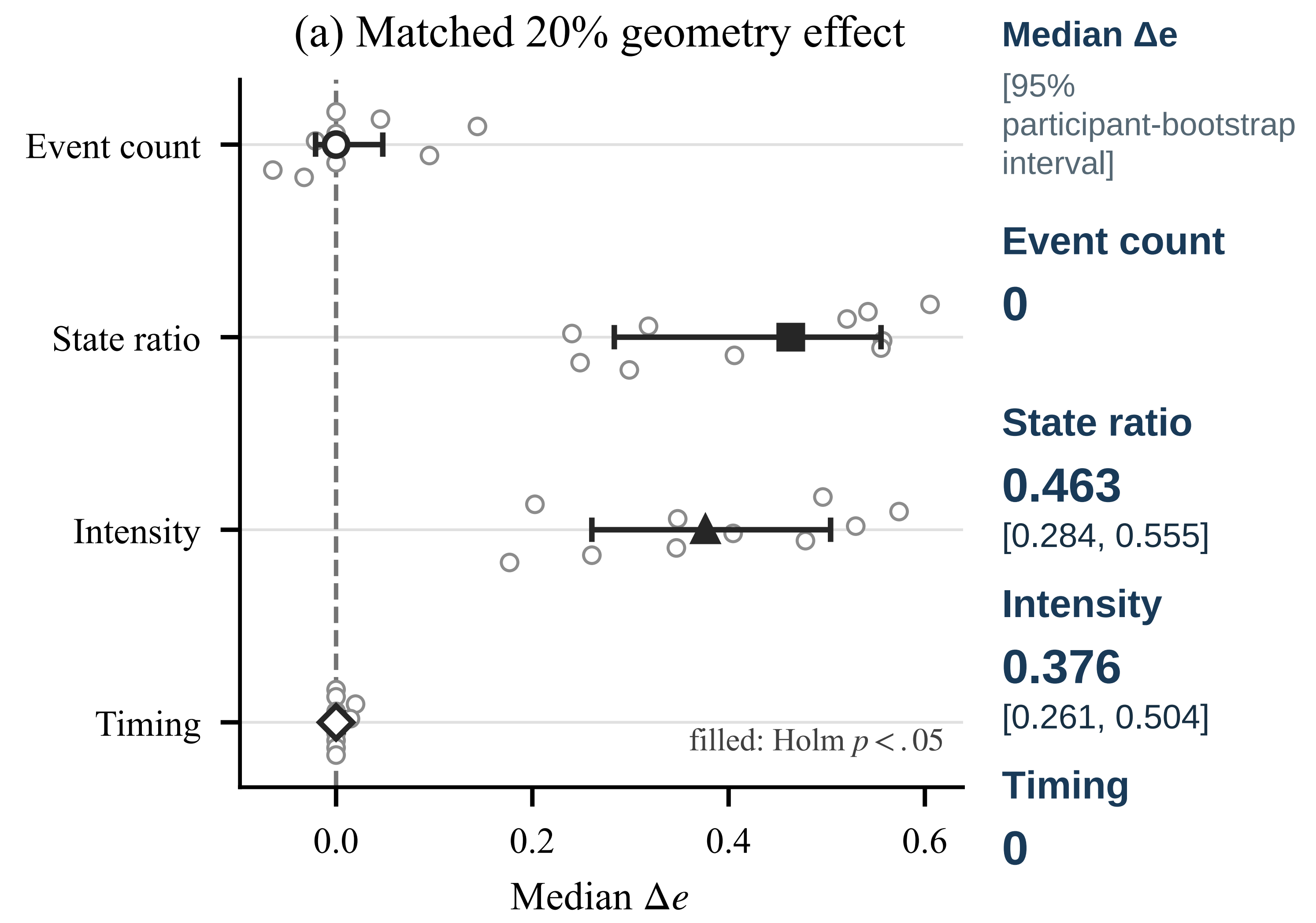


Fig. 1(a). Open circles: participants. Large markers: medians and 95% bootstrap intervals; filled: Holm  $p < .05$ .

State ratio and intensity: participant-level median  $\Delta e$  was positive in 10/10 participants for each. Holm-adjusted  $p = .0078$  for both.

Zero medians for event count and timing do not establish equivalence.

### Overall decision: inconclusive

Only two families had positive median  $\Delta e$ . This fails one necessary part of the frozen overall criterion: at least three families with positive medians.

## 5 Exploratory recovery

### Representative light-feature examples (n = 10)

20% time-window deletion; random deletion matches the actual valid-record count removed.

| Light                               | Contiguous 20% | Random 20%    |
|-------------------------------------|----------------|---------------|
| <b>Linear interpolation</b>         |                |               |
| Mobile                              | <b>+0.193</b>  | <b>−0.011</b> |
| Wearable                            | <b>+0.306</b>  | <b>−0.015</b> |
| <b>Same-time median<sup>1</sup></b> |                |               |
| Mobile                              | <b>+0.437</b>  | <b>−0.033</b> |
| Wearable                            | <b>+0.528</b>  | <b>−0.085</b> |

Gain =  $e(\text{no repair}) - e(\text{repair})$ . Positive gain means LOWER standardized error; negative gain means HIGHER standardized error.

Participant medians (n = 10), point estimates only. <sup>1</sup>Same-time median from the same participant’s other days.

Repairs use known deletion positions and intact other days offline; future observations may contribute. These are feature-recovery gains, not demonstrated prediction gains.